

Article

Digital Financial Services and Sustainable Development: Temporal Trade-Offs and the Moderating Role of Financial Literacy

Jihyung Han ¹  and Daekyun Ko ^{2,*} 

¹ Research Institute of Human Ecology, Seoul National University, Seoul 08826, Republic of Korea; ellyhan@snu.ac.kr

² Department of Consumer Science and Glocal Life-Care Convergence, Chungnam National University, Daejeon 34134, Republic of Korea

* Correspondence: dk.ko@cnu.ac.kr; Tel.: +82-42-821-6848

Abstract

Digital financial services have transformed consumer financial behavior, yet their effects on sustainable development outcomes remain poorly understood. This study examines how mobile financial services (MFS) usage influences financial behaviors across temporal dimensions and investigates the moderating role of financial literacy from a systemic sustainability perspective. Drawing on Construal Level Theory, Dual Process Theory, and Social Cognitive Theory, we analyze data from 21,757 U.S. adults from the 2021 National Financial Capability Study to explore relationships between MFS usage, financial literacy dimensions—objective knowledge (OK), subjective knowledge (SK), and perceived ability (PA)—and both short-term and long-term financial behaviors. The results reveal a dual temporal pattern: MFS usage negatively affects short-term behaviors, including spending control and emergency preparedness, while positively influencing long-term behaviors such as retirement planning and investment participation. Financial literacy dimensions demonstrate differential moderating effects, with OK providing protective benefits against short-term risks, while PA can paradoxically exacerbate these adverse short-term effects. These findings highlight complex implications for sustainable development, demonstrating how individual behaviors aggregate to influence systemic financial resilience and progress toward Sustainable Development Goals related to poverty reduction, economic growth, and inequality reduction. Policymakers should adopt behaviorally informed regulatory approaches that address temporal trade-offs. Educators should design digital-specific literacy programs emphasizing realistic risk assessment alongside confidence-building, thereby promoting sustainable financial behaviors in increasingly digital environments.

Keywords: digital financial services; mobile financial services; financial literacy; sustainable development; temporal trade-offs; financial behavior; financial resilience



Academic Editor: Seul Ki Lee

Received: 11 August 2025

Revised: 2 October 2025

Accepted: 9 October 2025

Published: 10 October 2025

Citation: Han, J.; Ko, D. Digital Financial Services and Sustainable Development: Temporal Trade-Offs and the Moderating Role of Financial Literacy. *Sustainability* **2025**, *17*, 8976. <https://doi.org/10.3390/su17208976>

Copyright: © 2025 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (<https://creativecommons.org/licenses/by/4.0/>).

1. Introduction

The rapid expansion of digital financial technologies has fundamentally transformed how individuals manage their financial resources, presenting both unprecedented opportunities and emerging challenges for sustainable economic development. Mobile Financial Services (MFS)—encompassing mobile payments, transfers, account management, and financial planning tools—have democratized access to financial services worldwide [1,2], contributing directly to key Sustainable Development Goals (SDGs), particularly SDG 1 (No

Poverty), SDG 8 (Decent Work and Economic Growth), and SDG 10 (Reduced Inequalities). These three goals were chosen because they directly map onto the behavioral mechanisms under study: short-term financial discipline links to poverty prevention (SDG 1), long-term financial planning supports economic growth through capital accumulation (SDG 8), and differential effects across financial literacy levels relate to inequality dynamics (SDG 10).

However, emerging evidence suggests the impact of MFS on sustainable financial behavior is multifaceted and potentially contradictory. While digital financial tools facilitate long-term financial planning through automated savings and investment platforms [3,4], the frictionless nature of digital transactions may inadvertently encourage impulsive spending behaviors and weaken short-term financial discipline [5]. Recent research demonstrates that mobile payment usage increases financial anxiety through overspending and overindebtedness [6], while theoretical work explains how mobile payment systems encourage spending by eliminating the psychological costs associated with traditional payment methods [7]. This dual nature raises a critical question: Do MFS ultimately support or undermine the behavioral foundations necessary for systemic financial resilience and sustainable economic development?

The relationship between individual financial behaviors and broader sustainable development extends beyond personal financial well-being to encompass systemic financial resilience and social equity. When individuals engage in sustainable financial behaviors—practices that support both immediate stability and long-term well-being—they contribute to household financial stability, which in turn supports the resilience of the financial system. Conversely, widespread individual financial instability can cascade into systemic risks that threaten economic sustainability and exacerbate inequalities.

Despite growing interest in digital financial services, three critical research gaps persist. First, existing research has examined MFS effects through fragmented lenses, focusing either on adoption patterns or isolated behavioral outcomes, without systematically investigating how digital financial services simultaneously affect different temporal dimensions of financial behavior [8,9]. Second, while studies typically examine short-term and long-term financial behaviors separately [10], none have empirically tested the theoretically predicted dual effects where digital convenience might undermine short-term discipline while enhancing long-term planning capabilities. Third, the moderating role of financial literacy's distinct dimensions remains insufficiently understood in digital contexts. Prior work suggests objective knowledge promotes prudent behaviors [10,11], whereas perceived ability may paradoxically increase risk-taking in frictionless digital environments [12]. Addressing these gaps is critical for individual well-being and systemic financial resilience; accordingly, this study provides one of the first large-scale examinations of how MFS usage influences sustainable financial behavior across temporal dimensions, while systematically investigating the moderating effects of financial literacy dimensions. Using data from 21,757 U.S. adults from the 2021 National Financial Capability Study, we analyze how MFS usage affects both short-term financial discipline and long-term financial planning, with financial literacy serving as a crucial moderating mechanism.

The research makes several important contributions. Drawing on established behavioral frameworks (Construal Level Theory (CLT), Dual Process Theory (DPT), and Social Cognitive Theory (SCT)), the study frames how digital environments alter decision-making, with details elaborated in the literature review. Empirically, it provides robust evidence of how digital financial services simultaneously affect different temporal dimensions of financial behavior. Practically, the findings offer actionable insights for developing behavior-informed financial regulations, designing digital financial literacy curricula, and creating fintech products that promote sustainable financial behaviors at both individual

and systemic levels. This study directly addresses the paradoxical question of whether MFS ultimately supports or undermines sustainable development objectives.

These contributions are critical for achieving individual financial well-being and broader economic sustainability in an increasingly digital financial landscape, particularly as policymakers seek to harness digital financial innovation for sustainable development while building resilient financial systems. While this analysis draws on U.S. survey data, it establishes an empirical foundation for future cross-cultural validation of digital finance's role in sustainable development.

2. Literature Review and Hypothesis Development

2.1. Theoretical Foundations and Sustainable Development Framework

The behavioral implications of digital financial services for sustainable development require an interdisciplinary theoretical approach that integrates psychological and economic perspectives. This study draws upon three complementary frameworks to explain how MFS influences individual financial behaviors with broader implications for achieving Sustainable Development Goals (SDGs), particularly SDG 1 (No Poverty), SDG 8 (Decent Work and Economic Growth), and SDG 10 (Reduced Inequalities).

CLT explains how psychological distance affects financial decision-making across temporal dimensions. Digital financial services reduce psychological friction and temporal distance, creating divergent effects: facilitating long-term financial planning while potentially undermining short-term discipline [13]. Recent evidence supports this dual pattern, showing that digital money reduces psychological distance and increases spending impulsivity [6,14].

DPT distinguishes between System 1 (automatic, intuitive) and System 2 (deliberate, analytical) decision-making processes [15]. MFS design inherently promotes System 1 processing through reduced transaction friction, while financial literacy serves as a cognitive resource enabling System 2 engagement [16].

SCT emphasizes self-regulation and self-efficacy in financial behavior [17]. Financial literacy dimensions—objective knowledge (OK), subjective knowledge (SK), and perceived ability (PA)—provide different self-regulatory resources, with potentially paradoxical effects in digital environments where overconfidence may undermine prudent decision-making [12].

Integrating CLT, DPT, and SCT, MFS reduces psychological distance and transaction frictions, tilting choices toward concrete, immediate payoffs while easing engagement with abstract, future-oriented goals. We therefore expect weaker short-term discipline (spending control, overdraft avoidance, emergency buffers) but stronger long-term planning (retirement preparation, investment participation), contingent on literacy resources that enable deliberation (OK) or inflate overconfidence (PA).

The selection of SDGs 1, 8, and 10 reflects the direct pathways through which digital financial behaviors aggregate to influence systemic sustainable development outcomes. SDG 1 (No Poverty) is related to individual financial behaviors through emergency preparedness, debt avoidance, and savings accumulation—behaviors that create household financial resilience and prevent poverty traps [18]. SDG 8 (Decent Work and Economic Growth) is linked to long-term financial planning, investment participation, and retirement savings, which contribute to capital market development and economic growth [19]. SDG 10 (Reduced Inequalities) is directly impacted by differential access to and outcomes from digital financial services, where an unequal distribution of financial literacy may exacerbate existing inequalities rather than mitigate them [20]. This framework recognizes that individual financial behaviors are not merely personal choices but building blocks of broader economic sustainability and social equity.

2.2. Digital Financial Services and Financial Behavior

Building on this theoretical foundation, empirical research has produced contradictory evidence regarding MFS effects on financial behavior, creating a digital financial paradox. Positive evidence demonstrates that MFS enhances financial inclusion by reducing transaction costs and expanding access to formal financial services, particularly for previously excluded populations [1]. Digital financial tools facilitate savings mobilization through automated features and lower barriers to account opening [3]. Long-term financial planning benefits from sophisticated budgeting applications, automated investment platforms, and accessible retirement planning tools once limited to wealthier users [4]. Studies in developing contexts show that digital financial services can significantly enhance household financial asset allocation and promote diversified investment portfolios [21,22].

However, emerging evidence reveals concerning behavioral implications that threaten short-term financial stability. Mobile payment usage increases financial anxiety through enhanced overspending and overindebtedness behaviors, with these effects serving as mediators between technology adoption and financial distress [6]. The theoretical foundation for these adverse effects lies in the “pain of paying” phenomenon—digital transactions eliminate psychological barriers that traditionally supported financial self-control by removing the tangible experience of monetary loss associated with cash payments [7]. Conceptually, this aligns with DPT’s shift toward automatic processing under reduced friction and with CLT’s contraction of psychological distance in immediate transactions. Mobile payments systematically promote impulsive buying behaviors through reduced transaction visibility, enhanced convenience, and weakened psychological resistance to spending [5,23].

This dual pattern creates significant implications for sustainable financial behavior and development outcomes. The temporal fault line is pivotal: the same features that automate saving and investing can, in daily money management, erode the frictions that anchor short-term financial discipline.

This study moves beyond adoption- or single-outcome-focused analyses by offering a temporally differentiated account that jointly models immediate (short-term financial discipline) and distant (long-term financial planning) behaviors under unified CLT-DPT-SCT premises. We examine post-adoption behavioral trade-offs, which differs from previous adoption-focused studies [8,24]. Prior research has examined temporal dimensions of financial behavior, but typically within more limited contexts. Financial literacy affects short-term and long-term behaviors differently across age groups [10]. Similar temporal distinctions emerge in the context of financial education interventions [4]. This line of inquiry extends to millennials, documenting how financial knowledge differentially influences immediate versus future-oriented financial decisions [11]. More recently, the nuanced role of financial confidence has been highlighted, with findings that overconfidence can undermine sustainable practices despite adequate knowledge [25]. While these studies enrich understanding of temporal patterns in financial behavior, they typically analyze demographic differences, educational effects, or confidence mechanisms within traditional financial contexts. Our approach tests simultaneous dual effects in a national sample and estimates moderation by multidimensional literacy (OK, SK, PA), extending beyond the single-outcome approaches used in prior research [3,4]. We extend beyond psychosocial determinants to post-adoption temporal trade-offs in a national sample, building upon but differentiating from previous work that focused on either adoption patterns or isolated outcome [24]. By contrast, we systemically model how digital financial services create opposing effects across temporal dimensions—undermining short-term discipline while enhancing long-term planning—within a unified theoretical framework that integrates CLT, DPT, and SCT. This temporal dimension is critical for understanding sustainable

development implications, as short-term financial instability can undermine long-term wealth accumulation essential for poverty reduction and economic resilience.

This body of research reveals three critical gaps. First, no studies have systematically examined the dual temporal effects theoretically predicted by CLT, where reduced psychological distance should create opposing effects on immediate versus distant financial decisions. Second, the moderating role of multidimensional financial literacy remains poorly understood, particularly given recent evidence that confidence-based literacy measures may have paradoxical effects in digital contexts. Third, the connection between individual behavioral changes and sustainable development outcomes has not been systematically theorized or empirically examined in digital financial contexts.

2.3. Financial Literacy in Digital Contexts

Financial literacy operates through fundamentally different mechanisms in digital environments compared to traditional financial contexts, with critical implications for sustainable development outcomes. The multidimensional nature of financial literacy—comprising OK, SK, and PA—creates differential effects that challenge conventional wisdom about the universally positive role of financial education.

OK reliably predicts superior portfolio choices and overall well-being [26], extends to digital protection against fraud [27], and even correlates with sustainable investing [25]. In digital contexts, OK functions as a System-2 enabler that counters frictionless-interface impulsivity, enabling more deliberative financial decision-making [16]. By contrast, SK is beneficial only when calibrated to actual competence—otherwise, overconfidence depresses risk-adjusted outcomes [28,29]. Most concerning, PA can turn counterproductive online: PA-driven overconfidence mediates links between MFS usage and high-cost borrowing and fosters premature adoption of complex products [12,30].

This confidence paradox has profound implications for sustainable development, as it suggests that traditional financial education approaches focused on building confidence may backfire in digital contexts without corresponding emphasis on realistic self-assessment and appropriate confidence calibration.

2.4. Research Gaps and Hypothesis Development

Three critical research gaps limit understanding of MFS effects on sustainable financial behavior and their implications for achieving key development goals. First, existing research examines MFS effects through fragmented theoretical lenses without systematically investigating how digital financial services simultaneously affect different temporal dimensions of financial behavior and their aggregate implications for systemic financial resilience [8,9]. Second, while studies examine short-term and long-term financial behaviors separately [10], none have empirically tested the theoretically predicted dual temporal effects where digital convenience creates fundamental trade-offs between immediate financial discipline and future-oriented financial planning capabilities. Third, the moderating role of financial literacy's distinct dimensions remains insufficiently understood in digital contexts, particularly given controversial findings where confidence-based literacy measures may paradoxically increase rather than decrease financial vulnerability [12,30].

These gaps are particularly problematic for sustainable development policy and practice, as they limit understanding of whether digital financial innovation supports or undermines the behavioral foundations necessary for achieving SDGs related to poverty reduction, economic growth, and inequality reduction. Without a systematic understanding of temporal trade-offs and conditional effects, policymakers cannot design interventions that maximize benefits while minimizing risks of digital financial inclusion initiatives. Given

the cross-sectional design, we acknowledge potential endogeneity concerns and address them through appropriate model specification and robustness checks (see Section 3).

2.4.1. Conceptual Framework

Figure 1 presents the integrated conceptual model positioning MFS usage as having dual effects varying by temporal dimensions, with financial literacy dimensions serving as crucial moderators that determine individual outcomes and their aggregate implications for sustainable development.

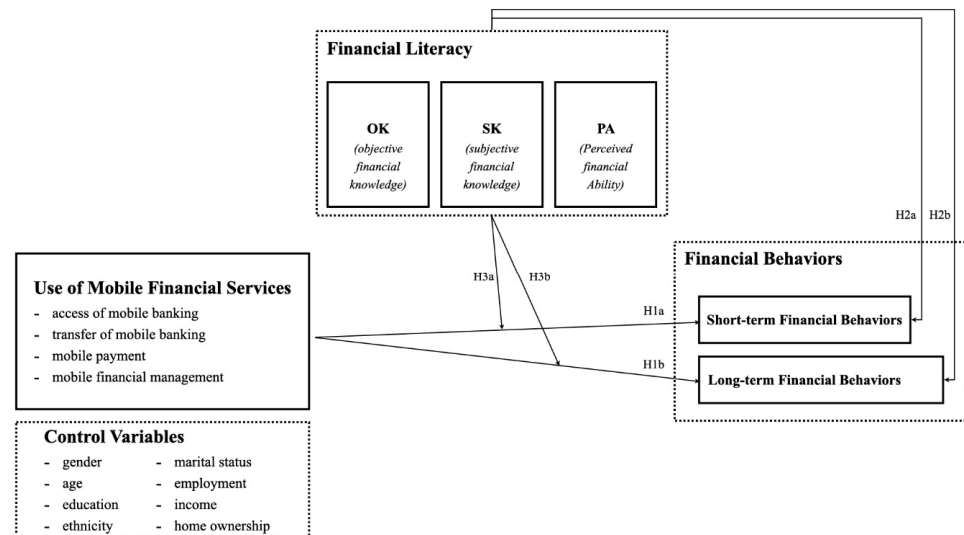


Figure 1. Conceptual model showing the relationships between MFS usage, financial literacy dimensions, and temporal financial behaviors.

2.4.2. Hypothesis Development

Drawing from the theoretical integration and empirical gaps identified:

H1: MFS usage produces temporal trade-offs in sustainable financial behavior, decreasing short-term financial discipline (H1a) while enhancing long-term financial planning (H1b), with implications for both individual financial stability and systemic financial resilience essential for sustainable development.

H2: Financial literacy dimensions positively influence both short-term (H2a) and long-term (H2b) sustainable financial behaviors, contributing to individual financial well-being and broader financial system stability necessary for sustainable economic development.

H3: Financial literacy dimensions moderate the relationship between MFS usage and financial behaviors, buffering adverse short-term effects (H3a) and enhancing positive long-term effects (H3b), thereby supporting pathways to systemic financial sustainability and inclusive economic development.

3. Materials and Methods

3.1. Dataset and Sample Selection

This study employs data from the 2021 National Financial Capability Study (NFCS) conducted by the FINRA Investor Education Foundation. The NFCS is a nationally representative survey designed to assess financial knowledge, behaviors, and the adoption of digital financial services among U.S. adults aged 18 and older. The dataset is extensively used and recognized in financial literacy and consumer behavior research, making it particularly suitable for this investigation [31,32].

The 2021 NFCS dataset contains 27,118 respondents collected through non-probability quota sampling aligned with demographic quotas derived from the U.S. Census Bureau's

American Community Survey (ACS), ensuring representativeness based on age, gender, race/ethnicity, education, and income. While this online panel sampling approach provides broad national coverage and efficient data collection, it may introduce potential biases related to digital access and internet literacy that could affect the measured representativeness of digital finance adoption. To ensure robust analyses and accurate inferences, respondents with missing values on key variables were removed using listwise deletion, resulting in a final analytical sample of 21,757 respondents.

To further enhance representativeness, national-level weights provided in the NFCS dataset (wgt_n2) were applied in all analyses, ensuring the results accurately reflect the demographic distribution of the broader U.S. adult population. The dataset is publicly available through the FINRA Foundation website (<https://www.finrafoundation.org/nfcs-data-and-downloads>) (accessed on 25 January 2025), enabling replication of this study's findings.

Table 1 summarizes descriptive statistics of the final analytical sample, providing insight into MFS usage rates, financial literacy levels, and demographic composition. Table 2 presents detailed distributions of short-term and long-term financial behaviors among respondents, enabling clear comparisons across financial behavior dimensions.

Table 1. Descriptive Statistics of Key Variables (N = 21,757).

	Frequency	Percentage	M (SD)
Use of MFS			
Access	16,317	75.0	
Transfer	12,004	55.2	
Payment	9449	43.4	
Management	9419	43.3	
Financial Literacy			
OK (0–6)			3.10 (1.67)
SK (1–7)			5.14 (1.30)
PA (1–4)			3.05 (0.87)
Age			
18–24	2303	10.6	
25–34	3675	16.9	
35–44	3401	15.6	
45–54	3665	16.8	
55–64	4100	18.8	
65+	4613	21.2	
Gender (reference group female)			
Male	10,779	49.5	
Female	10,978	50.5	
Ethnicity			
White	14,064	64.6	
Non-white	7693	35.4	
Education			
High school diploma or lower	5916	27.2	
Some college	6073	27.9	
Associate's degree	2558	11.8	
Bachelor's degree	4948	22.7	
Post graduate degree	2263	10.4	

Table 1. *Cont.*

	Frequency	Percentage	M (SD)
Marital status			
Married	10,765	49.5	
Single	7209	33.1	
Separated/divorced/widow	3783	17.4	
Dependent children			
No	14,381	66.1	
Yes	7377	33.9	
Employment			
Full-time employee	8586	39.5	
Self-employed	1714	7.9	
Part-time employee	1913	8.8	
Home maker	1341	6.2	
Student	626	2.9	
Permanently sick/disabled/or unable to work	1113	5.1	
Unemployed or temporarily laid off	1523	7.0	
Retired	4942	22.7	
Household income			
Less than \$15,000	2229	10.2	
\$15,000–\$24,999	2244	10.3	
\$25,000–\$34,999	2347	10.8	
\$35,000–\$49,999	3111	14.3	
\$50,000–\$74,999	4211	19.4	
\$75,000–\$99,999	2965	13.6	
\$100,000–\$149,999	2943	13.5	
\$150,000–\$199,999	1020	4.7	
\$200,000–\$299,999	469	2.2	
\$300,000 or more	218	1.0	
Home ownership (reference group no)			
No	8410	38.7	
Yes	13,347	61.3	

Notes: M = mean, SD = standard deviation (in parentheses); N = number of observations.

Table 2. Summary of Short-term and Long-term Behaviors (N = 21,757).

	Frequency	Percentage	M (SD)
Short-term behaviors index			2.38 (1.33)
Spending less than income	11,689	53.7	
No difficulty with bill payment	12,385	56.9	
No overdraft	16,986	78.1	
Emergency funds (ownership)	12,443	57.2	
Long-term behaviors index			2.27 (1.30)
Retirement planning (amount needed)	10,042	46.2	
Retirement account (ownership)	13,767	63.3	
Savings (ownership)	17,046	78.3	
Investments (ownership)	8497	39.1	

Notes: M = mean, SD = standard deviation (in parentheses); N = number of observations.

3.2. Variable Definitions

3.2.1. Dependent Variables: Sustainable Financial Behavior

Following prior frameworks that emphasize the temporal dimensions of financial behavior [10,11,31], this study assesses sustainable financial behavior using indices of both short-term and long-term financial behavior. These composite indices capture different dimensions of consumer financial actions that support both immediate financial stability and long-term economic well-being.

Short-Term Financial Behavior Index: This index measures immediate financial discipline and management strategies, including spending control, timely bill payment, overdraft avoidance, and maintaining emergency savings. Items are derived directly from the NFCS and consistent with established research [10,11]. Specifically, responsible behaviors (e.g., spending less than income, consistent bill payments, avoidance of overdrafts, and maintaining three-month emergency savings) are coded as 1, otherwise 0, resulting in an index ranging from 0 (least disciplined) to 4 (highly disciplined).

Long-Term Financial Behavior Index: This index evaluates future-oriented financial behaviors, including retirement planning, holding retirement accounts (such as 401(k) and IRA), savings accounts, and investments (stocks, bonds, and mutual funds). Each behavior is binary-coded (1 if engaged, zero otherwise), with total scores ranging from 0 (minimal long-term financial planning) to 4 (strong long-term financial planning). This approach builds upon prior studies that demonstrate the importance of distinguishing between short-term budgeting behaviors and long-term asset accumulation strategies [33,34].

3.2.2. Independent Variable: MFS Usage

MFS usage is measured through four distinct functionalities widely recognized in consumer financial technology research [1,35]:

- Access: Checking account balances via mobile apps
- Transfer: Sending money through mobile transactions
- Payment: Making in-store or online mobile payments
- Management: Budgeting or credit monitoring via mobile apps

Each functionality is binary-coded (0 = never used; 1 = used occasionally or frequently). These items are combined into a composite MFS Usage Index ranging from 0 to 4, where higher scores indicate greater engagement with MFS. This method reflects prior research that emphasizes the different functionalities of MFS may uniquely affect consumer financial outcomes [36].

3.2.3. Moderator: Financial Literacy Dimensions

This study conceptualizes financial literacy as multidimensional, encompassing three widely recognized components [37,38]:

- OK: Measured using six core financial concepts (interest compounding, inflation, bond pricing, time value of money, mortgage costs, and portfolio diversification), with scores ranging from 0 to 6
- SK: Self-assessed confidence in financial knowledge, measured on a 7-point Likert scale
- PA: Financial self-efficacy evaluated on a 4-point scale (1 = low confidence, 4 = high confidence)

This multifaceted approach aligns with contemporary research, which recognizes the complex interplay between objective financial knowledge, perceived confidence, and self-efficacy in shaping financial decisions and behaviors [28,39,40].

3.2.4. Control Variables

In alignment with previous studies [10,11], demographic and socioeconomic controls—including age, gender, education level, income, employment status, and homeownership—were included to account for confounding influences on financial behaviors. Incorporating these controls enhances the analytical rigor and validity of the findings by ensuring that observed relationships between MFS usage and financial behavior are not spurious or overly influenced by external demographic or socioeconomic factors.

3.3. Analytical Approach and Model Specification

3.3.1. Model Selection Rationale

Ordered logistic regression was selected for analyzing the financial behavior indices because the dependent variables (ranging from 0 to 4) represent ordinal outcomes with meaningful rank ordering. This approach appropriately models the cumulative probability structure while accounting for the ordered nature of behavioral intensities, unlike OLS regression, which assumes equal intervals between categories or multinomial logistic regression, ignoring ordering. For ordered logit, the proportional odds (parallel lines) assumption was evaluated via SPSS (version 24.0) Test of Parallel Lines (conceptually equivalent to the Brant test); results supported the specification.

Binary logistic regression was employed for individual financial behaviors, as these represent dichotomous outcomes that do not require the proportional odds assumption.

3.3.2. Formal Model Specifications

Phase 1: Main Effects Models (Ordered Logistic Regression)

For the financial behavior indices, the ordered logistic regression models are specified as:

$$\ln\left(\frac{P(Y_i \leq j)}{P(Y_i > j)}\right) = \alpha_j - \beta_1 MFS_i - \beta_2 OK_i - \beta_3 SK_i - \beta_4 PA_i - \gamma X_i$$

where

- Y_i represents the ordinal financial behavior index (short-term or long-term) for an individual i
- j presents the category threshold parameters (0, 1, 2, 3)
- α_j are the threshold parameters
- MFS_i is the MFS usage index
- OK_i, SK_i, PA_i are the three financial literacy dimensions
- X_i is the vector of control variables
- β and γ are the regression coefficients

Phase 2: Individual Behavior Models (Binary Logistic Regression)

For individual financial behaviors, the binary logistic regression models are specified as:

$$\ln\left(\frac{P(Y_i = 1)}{P(Y_i = 0)}\right) = \alpha + \beta_1 MFS_i + \beta_2 OK_i + \beta_3 SK_i + \beta_4 PA_i + \gamma X_i$$

where Y_i represents the binary outcome for each specific financial behavior.

Phase 3: Moderation Models

The moderation effects are examined through interaction terms in separate ordered logistic regression models:

$$\ln\left(\frac{P(Y_i \leq j)}{P(Y_i > j)}\right) = \alpha_j - \beta_1 MFS_i - \beta_2 FL_i - \beta_3 (MFS_i \times FL_i) - \gamma X_i$$

where FL_i represents each financial literacy dimension (OK, SK, or PA) in separate models, and β_3 captures the moderation effect.

3.3.3. Diagnostic Procedures

Multicollinearity Assessment: Variance inflation factors (VIF) were calculated for all predictor variables to assess potential multicollinearity issues. Most VIF values were below 3.0, with Age 65+ showing VIF = 5.4, reflecting natural correlation with retirement status and income variables, but remaining within acceptable bounds (<10). All other variables showed VIF values below 3.6, indicating no problematic collinearity among predictors (detailed results in Appendix A Table A1). The elevated VIF for the oldest age category is consistent with life-cycle research where age naturally correlates with retirement and accumulated wealth. However, threshold values of VIF need to be evaluated in the context of several other factors that influence the variance of regression coefficients, including sample size, model R^2 , and predictor variance. Values of VIF of 10, 20, 40, or even higher do not, by themselves, discount the results of regression analyses or necessitate variable elimination, use of ridge regression, or combining of independent variables into a single index [41]. There is no universal agreement on what VIF cut-off should be used, and any threshold should be based on practical considerations and evaluated against the overall model fit [42]. Our VIF diagnostics, combined with stable coefficient estimates across model specifications (Appendix A Table A6), confirm that multicollinearity does not compromise inference.

Model Fit Evaluation: Model performance was assessed using Nagelkerke R^2 , McFadden R^2 , -2 Log-Likelihood, and chi-square statistics. The proportional odds assumption for ordered logistic models was evaluated using SPSS Test of Parallel Lines. In contrast some models showed significant test statistics suggesting potential violations, ordered logistic regression was retained as the primary approach given its theoretical appropriateness for ordinal outcomes, with robustness verified through alternative specifications.

Marginal Effects and Visualization: Average marginal effects and interaction plots for significant moderating relationships were computed using supplementary routines consistent with standard implementations (see Section 4 and Appendix A Figure A1).

3.3.4. Analytical Implementation

A three-phase analytical approach was employed to systematically examine the relationships between MFS usage, financial literacy dimensions, and sustainable financial behaviors. All analyses were conducted using SPSS 30.0 supplemented with additional diagnostic routines, with national-level survey weights applied to ensure representativeness. Survey weights (wgt_n2) were applied throughout, and robust standard errors were computed to account for potential heteroscedasticity. Weighted and unweighted results were substantively equivalent in sign and significance (Appendix A Table A2).

Robustness Verification: To ensure the reliability of findings, several robustness checks were performed, including: (1) demographic subgroup analyses across age cohorts (18–34, 35–54, 55+) to assess consistency of dual effects across different populations and additional gender and income subgroup analyses (2) alternative model specifications comparing ordered logistic with OLS regression approaches to verify robustness to distributional assumptions, (3) weighted versus unweighted estimation comparisons to confirm representativeness is not driving results, and (4) variance inflation factor diagnostics to rule out multicollinearity concerns. These supplementary analyses confirm that the main findings are not sensitive to methodological choices (detailed results available in Appendix A Tables A1–A6).

This structured analytical framework aligns with best practices in empirical financial behavior research, thereby enhancing the interpretability and robustness of the findings [31,37].

3.4. Addressing Endogeneity Concerns

Given the cross-sectional nature of the data, potential endogeneity arising from reverse causality (individuals with better financial behaviors may be more likely to adopt MFS) and omitted variable bias represent a limitation. While instrumental variable approaches would provide stronger causal identification, appropriate instruments are not available in the NFCS dataset.

To address these concerns, we employ several strategies: (1) comprehensive control for observable confounders through demographic and socioeconomic variables, (2) model specification tests and diagnostic procedures, and (3) extensive robustness checks using alternative model specifications and subgroup analyses (detailed in the Appendix A). These strategies cannot entirely rule out endogeneity; panel or quasi-experimental designs would strengthen causal claims. However, the employed approaches provide confidence that the observed relationships are not driven entirely by reverse causality or significant omitted variables.

3.5. Data Availability and Ethics

This study uses publicly available 2021 NFCS data from the FINRA Foundation (<https://www.finrafoundation.org/knowledge-we-gain-share/nfcs>) (accessed on 25 January 2025). As this research involves secondary analysis of publicly accessible, de-identified data, no additional ethical approval was required. Detailed analytical procedures and variable construction methods are available upon request to ensure reproducibility. No generative AI tools were used in this research.

4. Results

4.1. Main Effects: Dual Temporal Patterns in MFS Usage (H1)

Table 3 presents the core findings examining how MFS usage affects financial behavior across temporal dimensions. The results provide strong empirical support for the hypothesized dual effects.

H1a: Short-term Financial Discipline (Supported): MFS usage consistently undermines short-term financial discipline across all functionalities (Table 3, Short-term index column). Mobile transfers and financial management tools indicate the most potent adverse effects (OR = 0.681 and 0.688, respectively, both $p < 0.001$). Relative to non-users of each function, these effects represent substantial reductions in the probability of maintaining high financial discipline ('high' = index 3–4). Mobile payments and access indicate moderate negative associations (OR = 0.775 and 0.824, both $p < 0.001$). This pattern supports the idea that digital convenience reduces psychological friction, traditionally supporting daily financial self-control.

H1b: Long-term Financial Planning (Supported): Conversely, three of four MFS functionalities enhance long-term financial behaviors (Table 3, Long-term index column). Financial management tools indicate the most potent positive effect (OR = 1.379, $p < 0.001$), representing substantial increases in the probability of high long-term planning engagement. Mobile transfers (OR = 1.190, $p < 0.001$) and mobile payments (OR = 1.129, $p < 0.001$) also indicate significant positive associations. Mobile access shows no statistically significant association with long-term behaviors. The dual nature is exemplified by financial management tools, which simultaneously increase the odds of long-term planning by 37.9% while decreasing the odds of short-term discipline by 31.2%.

Table 3. Ordered Logistic Regression Results on Short-term and Long-term Financial Behavior Index.

	Short-Term Behavior Index			Long-Term Behavior Index		
	Coefficient	SE	OR	Coefficient	SE	OR
Use of MFS						
Access	−0.194 ***	0.037	0.824	0.040	0.036	1.041
Transfer	−0.384 ***	0.033	0.681	0.174 ***	0.033	1.190
Payment	−0.253 ***	0.030	0.775	0.121 ***	0.030	1.129
Management	−0.374 ***	0.029	0.688	0.321 ***	0.029	1.379
Financial Literacy						
OK	0.140 ***	0.009	1.150	0.253 ***	0.009	1.288
SK	0.078 ***	0.011	1.081	0.165 ***	0.011	1.179
PA	0.940 ***	0.018	2.560	0.457 ***	0.017	1.579
Age group (ref: 18–24)						
25–34	−0.101	0.052	0.904	−0.065	0.053	0.937
35–44	−0.010	0.056	0.990	−0.158 **	0.057	0.854
45–54	0.172 **	0.057	1.188	−0.054	0.057	0.947
55–64	0.371 ***	0.061	1.449	0.296 ***	0.061	1.344
65+	0.506 ***	0.072	1.659	0.428 ***	0.072	1.534
Gender (ref: female)						
	−0.065 *	0.027	0.937	0.012	0.027	1.012
Ethnicity (ref: white)						
	0.046	0.030	1.047	0.000	0.030	1.000
Education (ref: high school diploma or lower)						
Some college	−0.071 *	0.035	0.931	0.356 ***	0.035	1.428
Associate’s degree	0.047	0.045	1.048	0.450 ***	0.045	1.568
Bachelor’s degree	0.251 ***	0.040	1.285	0.674 ***	0.040	1.962
Post graduate degree	0.081	0.052	1.084	0.782 ***	0.052	2.186
Marital status (ref: married)						
Single	0.120 **	0.037	1.127	−0.066	0.037	0.936
Separated/divorced/widow	−0.119 **	0.039	0.888	−0.333 **	0.039	0.717
Dependent children (ref: no)						
	−0.484 ***	0.031	0.616	0.013	0.031	1.013
Employment (ref: full-time employee)						
Self-employed	−0.105 *	0.050	0.900	−0.199 ***	0.050	0.820
Part-time employee	0.014	0.049	1.014	−0.284 ***	0.049	0.753
Home maker	−0.021	0.058	0.979	−0.466 ***	0.058	0.628
Student	0.023	0.082	1.023	−0.586 ***	0.083	0.557
Disabled or unable to work	−0.415 ***	0.065	0.660	−0.817 ***	0.066	0.442
Unemployed or temporarily laid off	−0.192 ***	0.055	0.825	−0.537 ***	0.056	0.584
Retired	0.045	0.050	1.046	−0.021	0.049	0.979
Household income (ref: less than \$15,000)						
\$15,000–\$24,999	−0.062	0.056	0.940	0.010	0.057	1.010
\$25,000–\$34,999	0.122 *	0.056	1.130	0.457 ***	0.057	1.579
\$35,000–\$49,999	0.229 ***	0.055	1.257	0.768 ***	0.056	2.155
\$50,000–\$74,999	0.623 ***	0.055	1.865	1.090 ***	0.056	2.974
\$75,000–\$99,999	0.866 ***	0.061	2.377	1.379 ***	0.062	3.971
\$100,000–\$149,999	1.015 ***	0.065	2.759	1.444 ***	0.065	4.238
\$150,000–\$199,999	1.099 ***	0.084	3.001	1.603 ***	0.083	4.968
\$200,000–\$299,999	1.510 ***	0.113	4.527	2.016 ***	0.112	7.508
\$300,000 or more	1.331 ***	0.151	3.785	1.943 ***	0.150	6.980
Home owner (ref: no)						
	0.259 ***	0.031	1.296	0.572 ***	0.031	1.772
Nagelkerke R ²	0.418			0.460		
McFadden R ²	0.163			0.184		
−2 Log-Likelihood	56,704.378			55,989.570		
Chi-square	11,081.458 ***			12,614.995 ***		

Notes: Significance level: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Regional fixed effects (State of residence) included.

4.2. Financial Literacy Main Effects (H2)

H2a & H2b: Consistently Positive Associations (Supported): All financial literacy dimensions indicate significant positive associations with both behavioral dimensions (Table 3). OK exhibits strong effects on both short-term ($OR = 1.150, p < 0.001$) and long-term ($OR = 1.288, p < 0.001$) behaviors. PA indicates the most potent effects ($OR = 2.560$ for short-term, $OR = 1.579$ for long-term, both $p < 0.001$), while SK ($OR = 1.124$ for short-term, $OR = 1.169$ for long-term, both $p < 0.001$) indicates modest but consistent benefits. These findings support that financial knowledge and confidence serve as crucial resources across temporal horizons.

4.3. Individual Behavior Analysis

Table 4 confirms the dual pattern at the individual behavior level. For short-term behaviors, overdraft avoidance shows the most severe deterioration with MFS usage across all functionalities. For long-term behaviors, financial management tools most strongly enhance retirement planning ($B = 0.347, p < 0.001$) and investment participation ($B = 0.199, p < 0.001$). Financial literacy dimensions maintain consistent positive associations across all individual behaviors.

Table 4. Binary Logistic Regression Results on Short- and Long-term Financial Behaviors.

	Short-Term Financial Behaviors								Long-Term Financial Behaviors							
	Spending		Bill Payment		No Overdraft		Emergency Funds		Retirement Planning		Retirement Account		Savings		Investments	
	B	SE	B	SE	B	SE	B	SE	B	SE	B	SE	B	SE	B	SE
Use of MFS																
Access	−0.100 *	0.041	−0.076	0.050	−0.277 ***	0.063	−0.330 ***	0.049	0.050	0.044	0.013	0.051	0.059	0.052	−0.024	0.047
Transfer	−0.134 ***	0.038	−0.401 ***	0.044	−0.542 ***	0.049	−0.309 ***	0.044	0.139 ***	0.041	0.021	0.047	0.130 **	0.047	0.170 ***	0.044
Payment	−0.163 ***	0.035	−0.354 ***	0.040	−0.399 ***	0.042	0.077	0.040	0.160 ***	0.038	0.049	0.043	−0.069	0.043	0.061	0.040
Management	−0.174 ***	0.034	−0.410 ***	0.038	−0.607 ***	0.040	−0.083 *	0.038	0.347 ***	0.036	0.172 ***	0.041	0.172 ***	0.041	0.199 ***	0.038
Financial Literacy																
OK	0.129 ***	0.010	0.072 ***	0.012	0.146 ***	0.013	0.095 ***	0.012	0.205 ***	0.011	0.217 ***	0.013	0.118 ***	0.013	0.246 ***	0.012
SK	0.003	0.013	0.065 ***	0.015	0.018	0.015	0.189 ***	0.015	0.217 ***	0.014	0.053 ***	0.015	0.054 ***	0.015	0.094 ***	0.015
PA	0.534 ***	0.020	1.171 ***	0.025	0.463 ***	0.024	0.950 ***	0.024	0.338 ***	0.021	0.377 ***	0.023	0.372 ***	0.022	0.280 ***	0.023
Nagelkerke R ²	0.173		0.439		0.242		0.413		0.286		0.475		0.235		0.337	
−2 Log-Likelihood	27,021.991		21,120.747		19,157.988		21,715.207		24,789.037		19,319.533		19,133.295		22,637.671	
Chi-square	3019.319 ***		8623.038 ***		3731.541 ***		7995.495 ***		5244.410 ***		9289.989 ***		3603.160 ***		6136.619 ***	

Notes: Significance level: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Regional fixed effects and control variables included for brevity.

4.4. Moderation Effects: The Confidence Paradox (H3)

Table 5 examines how financial literacy dimensions moderate MFS effects, revealing theoretically important differential patterns.

H3a: Short-term Moderation (Partially Supported): The moderation analysis reveals striking contrasts across literacy dimensions (Table 5). OK provides protective benefits ($B = 0.013, p < 0.05$), substantially buffering the adverse MFS effects. However, PA exhibits a significant adverse moderating effect ($B = −0.085, p < 0.001$), as shown in Table 5, indicating that overconfidence exacerbates rather than mitigates the adverse effects of MFS. High PA considerably amplifies the negative impact of MFS usage on short-term financial discipline (Figure A1, Panel B). This paradoxical finding supports that confidence undermines self-regulation in frictionless digital environments.

Table 5. Ordered Logistic Regression Results on Short- and Long-term Financial Behaviors Index.

	Short-Term Financial Behaviors						Long-Term Financial Behaviors					
	Model 1-1		Model 1-2		Model 1-3		Model 2-1		Model 2-2		Model 2-3	
	B	SE	B	SE	B	SE	B	SE	B	SE	B	SE
Mobile Financial Service Usage (MFU)	−0.343 ***	0.021	−0.258 ***	0.038	−0.050	0.034	0.206 ***	0.020	0.137 ***	0.038	0.232 ***	0.034
Financial Literacy												
OK	0.112 ***	0.015	0.139 ***	0.009	0.136 ***	0.130	0.279 ***	0.015	0.253 ***	0.009	0.252 ***	0.009
SK	0.078 ***	0.011	0.098 ***	0.020	0.079 ***	0.011	0.167 ***	0.011	0.154 ***	0.020	0.168 ***	0.011
PA	0.942 ***	0.018	0.939 ***	0.018	1.116 ***	0.029	0.456 ***	0.017	0.458 ***	0.017	0.500 ***	0.028
Interaction terms												
MFU × OK	0.013 *	0.006					−0.012 *	0.006				
MFU × SK			−0.009	0.007					0.006	0.007		
MFU × PA					−0.085 ***	0.011					−0.020	0.011
Nagelkerke R ²	0.417		0.417		0.419		0.459		0.459		0.459	
McFadden R ²	0.163		0.163		0.164		0.183		0.183		0.183	
−2 Log-Likelihood	56,712.245		56,715.655		56,654.810		56,020.072		56,024.389		56,021.143	
Chi-square	11,067.246 ***		11,063.837 ***		11,124.681		12,580.633 ***		12,576.317 ***		12,579.563 ***	

Notes: Significance level: * $p < 0.05$, *** $p < 0.001$. Regional fixed effects and control variables included for brevity.

H3b: Long-term Moderation (Limited Support): Long-term moderation effects are minimal (Table 5). Only the OK indicates a marginally significant negative interaction ($B = -0.012$, $p < 0.05$), suggesting that MFS benefits are slightly reduced among highly knowledgeable individuals. Neither SK nor PA significantly moderates long-term effects, indicating MFS benefits for future-oriented planning are robust across literacy levels.

4.5. Model Performance and Robustness

All models exhibit strong explanatory power (Nagelkerke $R^2 = 0.417$ – 0.460). Comprehensive robustness checks support stability: age-group analyses indicate consistent dual effects across demographics (Appendix A Table A3), gender and income subgroup analyses indicate similar patterns with some magnitude variations (Appendix A Tables A4 and A5), alternative OLS specifications yield identical patterns (Appendix A Table A6), and weighted versus unweighted analyses produce equivalent results (Appendix A Table A2). Multicollinearity diagnostics reveal no problematic issues (Appendix A Table A1). Interaction effects are visualized in Appendix A Figure A1, clearly illustrating the protective role of OK versus the risk-amplifying effect of PA.

5. Discussion

5.1. Summary of Key Findings and Theoretical Implications

This study examined the complex relationships between MFS usage, financial literacy dimensions, and sustainable financial behaviors using data from 21,757 U.S. adults. The findings reveal a fundamental paradox inherent in digital financial services: the same technological features that enhance long-term financial planning simultaneously undermine short-term financial discipline, with critical implications for sustainable development pathways.

5.1.1. Dual Effects and Critical Comparison with Existing Literature

Our findings provide strong empirical support for hypothesized dual temporal effects (H1), complicating optimistic accounts that claim MFS uniformly improves financial decision-making. Previous research has emphasized primarily positive effects of digital financial tools on savings behaviors, suggesting that technological convenience inherently promotes financial wellness [3,4]. However, our results reveal that this view captures only half the story. The negative associations between MFS usage and short-term financial discipline—including increased overdraft occurrences and weakened spending control—qualify the prevailing narrative in fintech literature that technological convenience inherently promotes financial wellness.

The magnitude of these dual effects, as demonstrated in Table 3, shows that management tools simultaneously enhance long-term planning while undermining short-term discipline, underscoring the temporal trade-offs. This pattern qualifies studies that emphasize short-term improvements via automation tools in digital financial services [8,24]. Our findings indicate instead that reduced psychological friction systematically undermines immediate financial self-control while facilitating long-term planning, supporting the “pain of paying” theory [7] over technological optimism approaches. This dual pattern creates what we term a “sustainability paradox,” where technologies that enhance long-term wealth building may simultaneously undermine the short-term stability essential for sustained economic resilience.

5.1.2. Financial Literacy Paradox in Digital Contexts

The study reveals that financial literacy operates fundamentally differently in digital environments than in traditional contexts, extending and challenging existing research. While all literacy dimensions showed positive main effects (H2), consistent with the patterns shown in Table 3, the moderation analysis (H3) revealed striking contrasts that contradict conventional wisdom. Most notably, PA paradoxically exacerbated rather than mitigated the adverse effects of MFS, directly opposing SCT predictions about self-efficacy [17].

This finding represents a significant departure from studies assuming that financial confidence uniformly benefits consumers [10,11]. Instead, our results align with emerging research on financial overconfidence [12,30], extending these findings to show that overconfidence systematically amplifies digital financial risks across multiple behavioral domains. The protective role of OK, as demonstrated in Table 5, confirms its continued importance while highlighting the insufficiency of confidence-based approaches to financial education in digital contexts.

5.1.3. Theoretical Integration and Contributions

This study successfully integrates CLT, DPT, and SCT to explain complex behavioral mechanisms underlying digital financial decision-making. The dual effects provide empirical support for CLT’s predictions about psychological distance [13], while the differential literacy effects advance DPT by revealing how knowledge versus confidence influences System 1 versus System 2 processing in digital environments [15]. The paradoxical effects of PA extend SCT by identifying conditions where self-efficacy becomes counterproductive.

These theoretical contributions bridge individual behavioral mechanisms with systemic sustainable development outcomes, demonstrating how digital financial behaviors aggregate to influence progress toward SDGs 1, 8, and 10. This integration provides a novel foundation for understanding the role of digital financial innovation in sustainable development.

5.2. Policy and Practical Implications

5.2.1. Behavioral Policy Interventions (Nudges)

The dual temporal effects suggest the need for differentiated behavioral interventions that address specific decision-making contexts. For short-term behaviors, policy should focus on introducing appropriate friction mechanisms—such as cooling-off periods for large transactions, spending alerts, and enhanced transaction visibility—that preserve accessibility while supporting self-regulation. The protective role of OK indicates that basic financial education initiatives focusing on factual knowledge remain crucial for digital financial safety.

For long-term behaviors, policies should remove barriers to beneficial financial planning tools while ensuring appropriate consumer protections. This includes streamlining access to retirement planning platforms and investment tools while maintaining strong disclosure requirements and fraud protections.

5.2.2. Structural Policy Interventions (Access Equity)

The differential effects of financial literacy dimensions highlight critical equity concerns requiring structural interventions. Patterns are broadly consistent across gender and income strata, with magnitude differences, indicating the value of targeted tailoring for vulnerable segments (Appendix A Tables A4 and A5). The paradoxical effects of PA suggest that traditional financial education approaches emphasizing confidence-building may be insufficient or even counterproductive in digital contexts. Instead, regulatory frameworks should require digital financial platforms to incorporate personalized risk assessment and adaptive interface design based on user literacy profiles.

Consumer protection measures should address the systematic vulnerabilities created by overconfidence, through requirements for enhanced disclosure to high-confidence users and mandatory safeguards for users exhibiting risk-amplifying behavioral patterns. These structural interventions are essential for ensuring digital financial innovation contributes to rather than undermines financial inclusion and social equity (SDG 10).

5.2.3. Industry and Educational Implications

Financial service providers should adopt user-centered design approaches that explicitly consider behavioral implications and sustainable development outcomes. This includes implementing behavioral safeguards such as customizable spending limits, automated savings features, and personalized educational content based on user risk profiles.

Educational programs require substantial modernization to address digital financial environments. Rather than focusing solely on confidence-building, curricula should emphasize realistic self-assessment, risk recognition, and the specific competencies required for safe digital financial navigation. Educational interventions should be tailored to different risk profiles while contributing to broader financial inclusion objectives.

While these implications emerge from U.S. data, the underlying behavioral mechanisms may extend to other institutional contexts, though cross-cultural validation would be essential for international policy application.

5.3. Limitations and Future Research Directions

5.3.1. Study Limitations

This study's cross-sectional design limits causal inference capabilities. The reliance on self-reported measures may introduce response biases, and the focus on U.S. adults limits generalizability across cultural contexts. While subgroup analyses by age, gender, and income were conducted, limitations in representativeness constrain reliable inference for racial/ethnic heterogeneity. Future research should examine potential racial/ethnic variations in MFS effects.

The temporal categorization of financial behaviors, while providing valuable insights, may not capture the full complexity of behavioral effects across different time horizons. Additionally, the study's financial literacy measures were developed primarily for traditional contexts and may not fully capture digital-specific competencies.

5.3.2. Future Research Priorities

Priority should be given to longitudinal studies tracking individuals over time as they adopt digital financial services, combined with experimental research manipulating specific platform features to establish causal relationships. Cross-cultural research examining how cultural factors influence digital financial service effects would enhance generalizability.

Research should examine vulnerable populations, including older adults and those with limited digital literacy, to understand how digital financial innovation affects financial

inclusion and exclusion. Additionally, studies should explore household-level dynamics and examine emerging technologies such as AI-driven financial management tools.

Methodological innovations combining survey data with transactional data would provide more comprehensive behavioral measures. Linking surveys with transactional traces will require robust privacy safeguards and ethical oversight. Policy research should rigorously evaluate different regulatory approaches and educational interventions to develop evidence-based frameworks supporting sustainable financial development in digital environments.

6. Conclusions

This study examined the impact of MFS on sustainable financial behavior across temporal dimensions and how different aspects of financial literacy influence these effects. The first hypothesis, that MFS usage weakens short-term financial discipline, was supported by the data, as evidenced by increased overspending and a higher likelihood of overdrafts. This confirms our prediction based on DPT that digital convenience triggers System 1 impulsivity. At the same time, our second hypothesis was also supported: MFS usage simultaneously promotes long-term financial planning behaviors, including retirement savings and investment participation. This validates our CLT framework, showing that digital tools facilitate abstract future-oriented thinking. Regarding our third hypothesis on the moderating role of financial literacy, we found partial support for this hypothesis. While all three dimensions—OK, SK, and PA—were positively associated with both behavioral domains, PA showed a paradoxical moderating effect, amplifying rather than mitigating the short-term risks of MFS.

Taken together, the findings offer several policy implications, each closely linked to the tested hypotheses. The demonstrated short-term vulnerabilities from our first hypothesis necessitate regulatory interventions that reintroduce friction into digital transactions—for example, mandatory spending alerts or enhanced transaction visibility to counter impulsive behavior. Conversely, the positive long-term effects confirmed by our second hypothesis justify policy initiatives to expand MFS-based access to retirement and investment tools, particularly for underserved populations. The paradoxical role of PA, as revealed by our third hypothesis, suggests that financial education programs must strike a balance between building confidence and providing realistic self-assessment. Meanwhile, fintech providers should design adaptive safeguards that protect overconfident users without undermining their autonomy.

This research makes three primary contributions to the literature. Empirically, it provides one of the first large-scale examinations ($N > 21,000$) of MFS's dual temporal effects on financial behavior. Theoretically, it integrates CLT, DPT, and SCT to demonstrate how digital convenience fundamentally alters intertemporal decision-making through both knowledge-based protection and confidence-based risk. Practically, it offers actionable insights for policymakers, educators, and industry stakeholders seeking to harness digital financial innovation for sustainable development while mitigating unintended risks.

Several limitations should be acknowledged. The cross-sectional design constrains causal inference, and future research employing longitudinal panel designs or randomized controlled trials would help establish temporal precedence. Our reliance on self-reported behaviors may introduce social desirability bias or recall errors, suggesting that future studies should incorporate objective transactional data to complement survey measures. The sample focuses on U.S. adults, which may limit generalizability to contexts with different cultural norms or regulatory environments. Additionally, our financial literacy measures were developed for traditional contexts. They may not fully capture digital-specific competencies, pointing to the need for new instruments tailored to the digital finance environment.

Building on these findings and limitations, future research should pursue several directions. Cross-national comparative studies would help test the contextual boundary conditions of our framework. At the same time, the development of digital-specific financial literacy instruments could better capture competencies unique to mobile financial environments. Examining heterogeneous effects across demographic subgroups, particularly vulnerable populations, would also enhance our understanding of who benefits from and who is harmed by the expansion of mobile financial services.

Author Contributions: Conceptualization, J.H.; methodology, J.H. and D.K.; software, J.H.; validation, J.H. and D.K.; formal analysis, J.H.; investigation, J.H.; resources, J.H.; data curation, J.H.; writing—original draft preparation, J.H. and D.K.; writing—review and editing, J.H. and D.K.; visualization, D.K.; supervision, J.H.; project administration, J.H. All authors have read and agreed to the published version of the manuscript.

Funding: This work was supported by the Ministry of Education of the Republic of Korea and the National Research Foundation of Korea (NRF-2022S1A5B5A16056267).

Institutional Review Board Statement: Ethical review and approval were waived for this study as it involved secondary analysis of publicly available, de-identified data from the 2021 National Financial Capability Study. The original data collection by the FINRA Investor Education Foundation was conducted in accordance with appropriate ethical standards. No additional ethical approval was required for this secondary data analysis under institutional guidelines for research using publicly available, de-identified datasets.

Informed Consent Statement: This study utilized publicly available secondary data from the 2021 National Financial Capability Study conducted by the FINRA Investor Education Foundation. Informed consent was obtained by the original data collectors from all participants involved in the original study. No additional informed consent was required for this secondary data analysis as the data are publicly available and de-identified.

Data Availability Statement: The original data presented in the study are openly available in the FINRA Foundations National Financial Capability Study repository at <https://www.finrafoundation.org/national-financial-capability-study> (accessed on 25 January 2025).

Acknowledgments: The authors acknowledge the FINRA Investor Education Foundation for making the National Financial Capability Study data publicly available for research purposes. During the preparation of this manuscript, the authors used generative artificial intelligence tools for English language editing and revision purposes only. All research design, data analysis, interpretation of results, and content development were conducted entirely by the authors. The authors have reviewed and edited all AI-assisted language revisions and take full responsibility for the content of this publication.

Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

MFS	Mobile Financial Services
OK	Objective Knowledge
SK	Subjective Knowledge
PA	Perceived Ability
CLT	Construal Level Theory
DPT	Dual Process Theory
SCT	Social Cognitive Theory

Appendix A

Table A1. Variance Inflation Factor (VIF) Diagnostics.

Variable	VIF
Use of MFS	
Access	1.506
Transfer	1.711
Payment	1.432
Management	1.306
Financial Literacy	
OK	1.371
SK	1.300
PA	1.350
Age group (ref: 18–24)	
25–34	2.498
35–44	2.703
45–54	2.971
55–64	3.644
65+	5.440
Gender (ref: female)	1.171
Ethnicity (ref: white)	1.118
Education (ref: high school diploma or lower)	
Some college	1.540
Associate’s degree	1.344
Bachelor’s degree	1.741
Post graduate degree	1.546
Marital status (ref: married)	
Single	1.922
Separated/divorced/widow	1.374
Dependent children (ref: no)	1.391
Employment (ref: full-time employee)	
Self-employed	1.143
Part-time employee	1.223
Home maker	1.239
Student	1.244
Disabled or unable to work	1.296
Unemployed or temporarily laid off	1.281
Retired	2.591
Household income (ref: less than \$15,000)	
\$15,000–\$24,999	1.872
\$25,000–\$34,999	1.992
\$35,000–\$49,999	2.393
\$50,000–\$74,999	3.023
\$75,000–\$99,999	2.765
\$100,000–\$149,999	3.026
\$150,000–\$199,999	1.856
\$200,000–\$299,999	1.429
\$300,000 or more	1.203
Home owner (ref: no)	1.446

Notes: Age 65+ (VIF = 5.4) reflects natural correlation with retirement status and income; all other variables < 3.6.

Table A2. Key Variables Comparison: Weighted vs. Unweighted Results.

	Short-Term Behavior Index				Long-Term Behavior Index			
	Weighted		Unweighted		Weighted		Unweighted	
	Coef.	<i>p</i>	Coef.	<i>p</i>	Coef.	<i>p</i>	Coef.	<i>p</i>
Use of MFS								
Access	−0.194 ***	<0.001	−0.237 ***	<0.001	0.040	0.261	0.042	0.238
Transfer	−0.384 ***	<0.001	−0.340 ***	<0.001	0.174 ***	<0.001	0.176 ***	<0.001
Payment	−0.253 ***	<0.001	−0.243 ***	<0.001	0.121 ***	<0.001	0.109 ***	<0.001
Management	−0.374 ***	<0.001	−0.394 ***	<0.001	0.321 ***	<0.001	0.316 ***	<0.001
Financial Literacy								
OK	0.140 ***	<0.001	0.145 ***	<0.001	0.253 ***	<0.001	0.251 ***	<0.001
SK	0.078 ***	<0.001	0.073 ***	<0.001	0.165 ***	<0.001	0.167 ***	<0.001
PA	0.940 ***	<0.001	0.970 ***	<0.001	0.457 ***	<0.001	0.462 ***	<0.001

Notes: *** $p < 0.001$. Results substantively equivalent across weighting schemes. Dual pattern (negative short-term, positive long-term MFS effects) robust.

Table A3. Age-Group Subsamples Analysis.

	18–34 Years (N = 5978)		35–54 Years (N = 7066)		55+ Years (N = 8713)	
	Coef.	<i>p</i>	Coef.	<i>p</i>	Coef.	<i>p</i>
Short-term behavior Index						
Use of MFS						
Access	−0.223 *	0.026	−0.289 ***	<0.001	−0.234 ***	<0.001
Transfer	−0.461 ***	<0.001	−0.447 ***	<0.001	−0.253 ***	<0.001
Payment	−0.318 ***	<0.001	−0.221 ***	<0.001	−0.169 **	0.002
Management	−0.365 ***	<0.001	−0.391 ***	<0.001	−0.263 ***	<0.001
Financial Literacy						
OK	0.134 ***	<0.001	0.151 ***	<0.001	0.113 ***	<0.001
SK	0.105 ***	<0.001	0.065 ***	<0.001	0.094 ***	<0.001
PA	0.709 ***	<0.001	0.893 ***	<0.001	1.157 ***	<0.001
Model Fit Statistics						
Nagelkerke R ²	0.275		0.374		0.433	
Long-term behavior Index						
Use of MFS						
Access	−0.042	0.674	0.147 *	0.038	0.070	0.143
Transfer	0.342 ***	<0.001	0.162 **	0.003	0.090	0.097
Payment	0.114 *	0.040	0.127 *	0.012	0.088	0.104
Management	0.444 ***	<0.001	0.430 ***	<0.001	0.145 **	0.004
Financial Literacy						
OK	0.241 ***	<0.001	0.255 ***	<0.001	0.253 ***	<0.001
SK	0.186 ***	<0.001	0.160 ***	<0.001	0.137 ***	<0.001
PA	0.278 ***	<0.001	0.392 ***	<0.001	0.599 ***	<0.001
Model Fit Statistics						
Nagelkerke R ²	0.381		0.479		0.478	

Notes: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. All models include full set of demographic and socioeconomic controls. Results generally support the dual pattern of MFS effects across age groups, with some variations in effect magnitude, particularly for long-term behaviors in older adults.

Table A4. Robustness Check: Gender Subgroup Analysis of MFS Effects on Financial Behaviors.

	Male (N= 10,779)		Female (N = 10,978)	
	Coef.	<i>p</i>	Coef.	<i>p</i>
Short-term behavior Index				
Use of MFS				
Access	−0.226 ***	<0.001	−0.179 ***	<0.001
Transfer	−0.334 ***	<0.001	−0.432 ***	<0.001
Payment	−0.300 ***	<0.001	−0.199 ***	<0.001
Management	−0.331 ***	<0.001	−0.401 ***	<0.001
Financial Literacy				
OK	0.184 ***	<0.001	0.091 ***	<0.001
SK	0.078 ***	<0.001	0.087 ***	<0.001
PA	0.868 ***	<0.001	1.008 ***	<0.001
Model Fit Statistics				
Nagelkerke R ²	0.401		0.438	
Long-term behavior Index				
Use of MFS				
Access	0.042	0.418	0.036	0.468
Transfer	0.110 *	0.023	0.213 ***	<0.001
Payment	0.096 *	0.033	0.126 **	0.003
Management	0.427 ***	<0.001	0.227 ***	<0.001
Financial Literacy				
OK	0.266 ***	<0.001	0.251 ***	<0.001
SK	0.198 ***	<0.001	0.142 ***	<0.001
PA	0.412 ***	<0.001	0.493 ***	<0.001
Model Fit Statistics				
Nagelkerke R ²	0.440		0.466	

Notes: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. All models include full set of demographic and socioeconomic controls. Results confirm the dual pattern of MFS effects across gender groups. Notable differences include stronger negative short-term effects of transfers and management tools among females, and stronger positive long-term effects of transfers among females, while management tools show stronger positive effects among males.

Table A5. Robustness Check: Income Subgroup Analysis of MFS Effects on Financial Behaviors.

	Low Income (<\$35k) (N = 4035)		Middle Income (\$35–\$74k) (N = 3776)		High Income (≥\$75k) (N = 3167)	
	Coef.	<i>p</i>	Coef.	<i>p</i>	Coef.	<i>p</i>
Short-term behavior Index						
Use of MFS						
Access	−0.213 ***	<0.001	−0.244 ***	<0.001	−0.154 *	0.027
Transfer	−0.520 ***	<0.001	−0.356 ***	<0.001	−0.299 ***	<0.001
Payment	−0.191 ***	<0.001	−0.148 **	0.005	−0.385 ***	<0.001
Management	−0.242 ***	<0.001	−0.355 ***	<0.001	−0.496 ***	<0.001
Financial Literacy						
OK	0.030	0.055	0.104 ***	<0.001	0.258 ***	<0.001
SK	0.140 ***	<0.001	0.118 ***	<0.001	−0.052 *	0.022
PA	0.782 ***	<0.001	1.079 ***	<0.001	1.049 ***	<0.001
Model Fit Statistics						
Nagelkerke R ²	0.294		0.382		0.346	
Long-term behavior Index						

Table A5. Cont.

	Low Income (<\$35k) (N = 4035)		Middle Income (\$35–\$74k) (N = 3776)		High Income (>=\$75k) (N = 3167)	
	Coef.	<i>p</i>	Coef.	<i>p</i>	Coef.	<i>p</i>
Use of MFS						
Access	0.070	0.281	0.011	0.854	0.004	0.948
Transfer	0.246 ***	<0.001	0.180 ***	0.001	0.108	0.066
Payment	0.044	0.406	0.111 *	0.033	0.195 ***	<0.001
Management	0.401 ***	<0.001	0.324 ***	<0.001	0.317 ***	<0.001
Financial Literacy						
OK	0.206 ***	<0.001	0.268 ***	<0.001	0.277 ***	<0.001
SK	0.120 ***	<0.001	0.163 ***	<0.001	0.271 ***	<0.001
PA	0.345 ***	<0.001	0.494 ***	<0.001	0.499 ***	<0.001
Model Fit Statistics						
Nagelkerke R ²	0.270		0.299		0.282	

Notes: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. All models include full set of demographic and socioeconomic controls. Results support the dual pattern of MFS effects across income groups with notable variations. High-income groups show stronger negative effects for payment and management tools on short-term behaviors, while low-income groups demonstrate stronger positive effects for transfer services on long-term behaviors. OK effects strengthen with income level, while SK shows mixed patterns across income strata.

Table A6. Alternative Model Specifications: OLS vs. Ordered Logistic Regression.

	Short-Term Behavior Index				Long-Term Behavior Index			
	OLS		Ordered Logit		OLS		Ordered Logit	
	Coef.	<i>p</i>	Coef.	<i>p</i>	Coef.	<i>p</i>	Coef.	<i>p</i>
Use of MFS								
Access	−0.112 ***	<0.001	−0.194 ***	<0.001	0.006	0.742	0.040	0.261
Transfer	−0.216 ***	<0.001	−0.384 ***	<0.001	0.086 ***	<0.001	0.174 ***	<0.001
Payment	−0.135 ***	<0.001	−0.253 ***	<0.001	0.059 ***	<0.001	0.121 ***	<0.001
Management	−0.210 ***	<0.001	−0.374 ***	<0.001	0.173 ***	<0.001	0.321 ***	<0.001
Financial Literacy								
OK	0.075 ***	<0.001	0.140 ***	<0.001	0.142 ***	<0.001	0.253 ***	<0.001
SK	0.046 ***	<0.001	0.078 ***	<0.001	0.083 ***	<0.001	0.165 ***	<0.001
PA	0.543 ***	<0.001	0.940 ***	<0.001	0.251 ***	<0.001	0.457 ***	<0.001
Model Fit Statistics								
R ² /Nagelkerke R ²	0.407		0.418		0.450		0.460	
Adjusted R ² /McFadden R ²	0.406		0.163		0.449		0.184	

Notes: *** $p < 0.001$. All models include full demographic and socioeconomic controls. OLS and ordered logistic results show identical coefficient directions and significance patterns, confirming robustness to model specification choice. Coefficient magnitudes differ due to different scaling but substantive conclusions unchanged. The dual pattern of MFS effects (negative short-term, positive long-term) is robust across both analytical approaches.

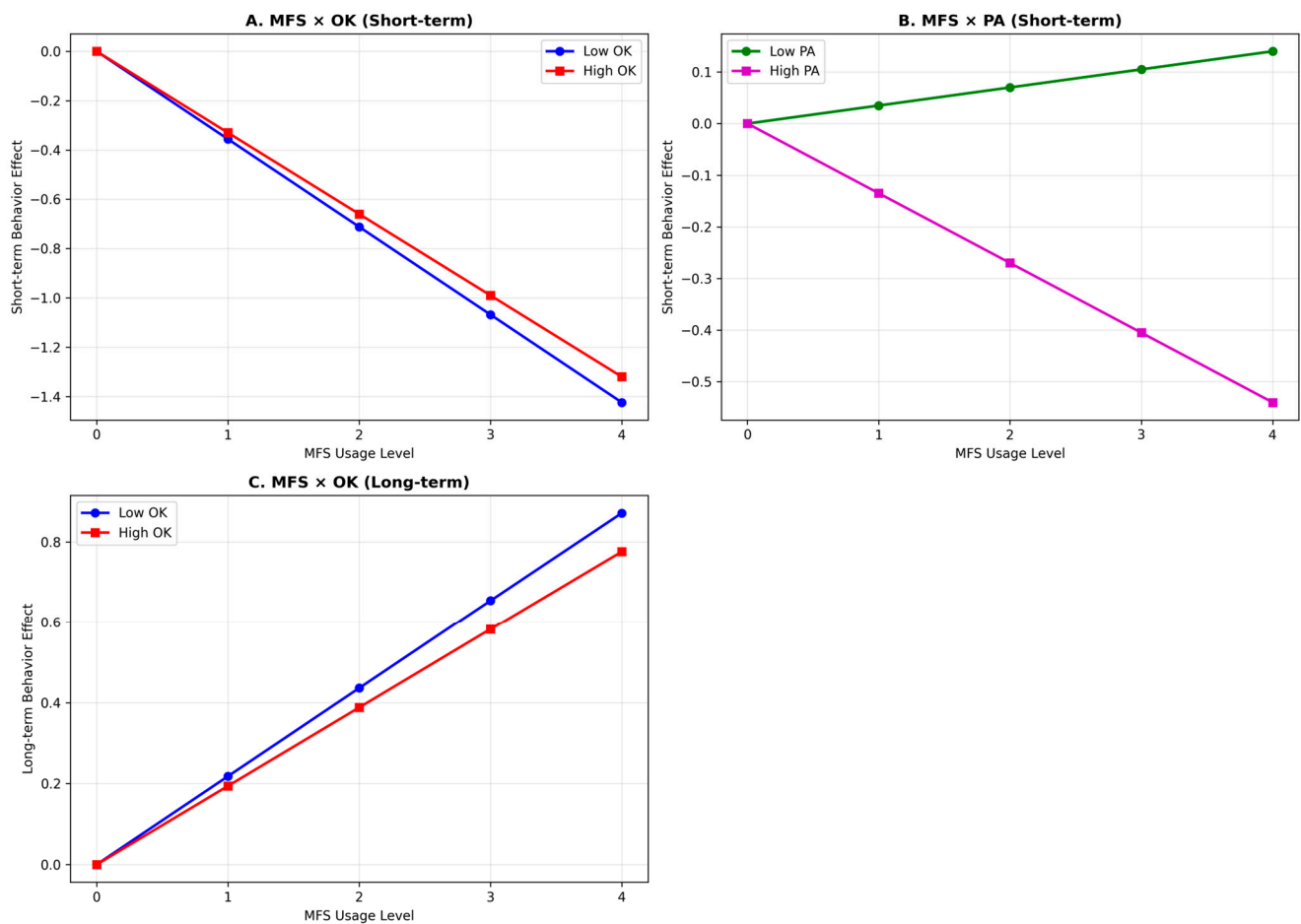


Figure A1. Interaction Effects of MFS Usage and Financial Literacy Dimensions on Financial Behaviors. (**Panel A**): MFS × OK (Short-term Behaviors). (**Panel B**): MFS × PA (Short-term Behaviors). (**Panel C**): MFS × OK (Long-term Behaviors).

References

1. Shaikh, A.A.; Alamoudi, H.; Alharthi, M.; Glavee-Geo, R. Advances in mobile financial services: A review of the literature and future research directions. *Int. J. Bank Mark.* **2023**, *41*, 1–33. [\[CrossRef\]](#)
2. Sharma, H.; Díaz Andrade, A. Digital financial services and human development: Current landscape and research prospects. *Inf. Technol. Dev.* **2023**, *29*, 582–606. [\[CrossRef\]](#)
3. Ouma, S.A.; Odongo, T.M.; Were, M. Mobile financial services and financial inclusion: Is it a boon for savings mobilization? *Rev. Dev. Financ.* **2017**, *7*, 29–35. [\[CrossRef\]](#)
4. Wagner, J.; Walstad, W.B. The effects of financial education on short-term and long-term financial behaviors. *J. Consum. Aff.* **2019**, *53*, 234–259. [\[CrossRef\]](#)
5. Zhang, L.; Zhao, H. From excessive spending to debt delinquency: Should we blame mobile payments? *Comput. Hum. Behav.* **2025**, *165*, 108533. [\[CrossRef\]](#)
6. An, T.; Xiao, J.J.; Porto, N.; Cruz, L. Mobile payment, financial behavior and financial anxiety: A multi-group structural equation modeling study. *Int. J. Bank Mark.* **2025**, *43*, 549–568. [\[CrossRef\]](#)
7. Alkadash, T.M.; Ateeq, A.; Dawwas, M.; Ibrahim, S.B.; Alkadash, Y.M.; Alkadash, R.M.; Alhassan, T. Exploring the psychological and behavioral effects of mobile payment systems on consumer spending: A theoretical perspective. In *Tech Fusion in Business and Society*; Hamdan, R.K., Ed.; Springer: Cham, Switzerland, 2025; Volume 234. [\[CrossRef\]](#)
8. Bashir, B.K.; Muhammad, T. Factors influencing the adoption of mobile financial services in enhancing financial inclusion: Using structural equation modeling for the decision-making approach. *Innovare J. Soc. Sci.* **2023**, *11*, 19–27. [\[CrossRef\]](#)
9. Mora, T.; Prior, F. The impact of mobile financial services' usage on microfinance delinquency. *Appl. Econ.* **2018**, *50*, 5354–5365. [\[CrossRef\]](#)
10. Henager, R.; Cude, B.J. Financial literacy and long- and short-term financial behavior in different age groups. *J. Financ. Couns. Plan.* **2016**, *27*, 3–19. [\[CrossRef\]](#)

11. Kim, K.T.; Anderson, S.G.; Seay, M.C. Financial knowledge and short-term and long-term financial behavior of millennials in the United States. *J. Fam. Econ. Issues* **2018**, *40*, 194–208. [\[CrossRef\]](#)
12. Mokhtari, M.; Chawla, I. Financial literacy overconfidence, mobile financial service use, and high-cost borrowing. *J. Financ. Couns. Plan.* **2023**, *34*, 184–203. [\[CrossRef\]](#)
13. Trope, Y.; Liberman, N. Construal-level theory of psychological distance. *Psychol. Rev.* **2010**, *117*, 440–463. [\[CrossRef\]](#)
14. Zhou, K.; Ye, J.; Liu, X.X. Is cash perceived as more valuable than digital money? The mediating effect of psychological ownership and psychological distance. *Mark. Lett.* **2023**, *34*, 55–68. [\[CrossRef\]](#)
15. Kahneman, D. A perspective on judgment and choice: Mapping bounded rationality. *Am. Psychol.* **2003**, *58*, 697–720. [\[CrossRef\]](#)
16. Kumar, P.; Islam, M.A.; Pillai, R.; Sharif, T. Analysing the behavioural, psychological, and demographic determinants of financial decision making of household investors. *Heliyon* **2023**, *9*, e13085. [\[CrossRef\]](#)
17. Bandura, A. Human agency in social cognitive theory. *Am. Psychol.* **1989**, *44*, 1175–1184. [\[CrossRef\]](#)
18. Ozili, P.K. Financial inclusion research around the world: A review. *Forum Soc. Econ.* **2021**, *50*, 457–479. [\[CrossRef\]](#)
19. Zehri, C.; El Amin, B.M.; Kadja, A.; Inaam, Z.; Sekrafi, H. Exploring the nexus of decent work, financial inclusion, and economic growth: A study aligned with SDG 8. *Sustain. Futures* **2024**, *7*, 100213. [\[CrossRef\]](#)
20. Dash, A.; Mohanta, G. Fostering financial inclusion for attaining sustainable goals: What contributes more to the inclusive financial behaviour of rural household in India? *J. Clean. Prod.* **2024**, *449*, 141731. [\[CrossRef\]](#)
21. Li, K.; Mengmeng, H.; Huo, J. Digital inclusive finance and asset allocation of Chinese residents: Evidence from the China household finance survey. *PLoS ONE* **2022**, *17*, e0267055. [\[CrossRef\]](#) [\[PubMed\]](#)
22. Dong, J. Digital finance's impact on household portfolio diversity: Evidence from Chinese households. *Financ. Res. Lett.* **2024**, *70*, 106347. [\[CrossRef\]](#)
23. Faraz, N.; Anjum, A. Spendception: The psychological impact of digital payments on consumer purchase behavior and impulse buying. *Behav. Sci.* **2025**, *15*, 387. [\[CrossRef\]](#)
24. Goyal, K.; Kumar, S.; Hoffmann, A. The direct and indirect effects of financial socialization and psychological characteristics on young professionals' personal financial management behavior. *Int. J. Bank Mark.* **2023**, *41*, 1550–1584. [\[CrossRef\]](#)
25. Aristei, D.; Gallo, M. Financial knowledge, confidence, and sustainable financial behavior. *Sustainability* **2021**, *13*, 10926. [\[CrossRef\]](#)
26. Chu, Z.; Wang, Z.; Xiao, J.J.; Zhang, W. Financial literacy, portfolio choice and financial well-being. *Soc. Indic. Res.* **2017**, *132*, 799–820. [\[CrossRef\]](#)
27. Isaia, E.; Oggero, N.; Sandretto, D. Is financial literacy a protection tool from online fraud in the digital era? *J. Behav. Exp. Financ.* **2024**, *44*, 100977. [\[CrossRef\]](#)
28. Chen, B.; Giannikos, C.I.; Lou, J. Is more financial literacy always beneficial? An investigation through a mediator. *J. Risk Financ. Manag.* **2023**, *16*, 53. [\[CrossRef\]](#)
29. Grežo, M. Overconfidence and financial decision-making: A meta-analysis. *Rev. Behav. Financ.* **2021**, *13*, 276–296. [\[CrossRef\]](#)
30. Vörös, Z.; Szabó, Z.; Kehl, D.; Kovács, O.B.; Papp, T.; Schepp, Z. The forms of financial literacy overconfidence and their role in financial well-being. *Int. J. Consum. Stud.* **2021**, *45*, 1292–1308. [\[CrossRef\]](#)
31. Lusardi, A.; Streeter, J.L. Financial literacy and financial well-being: Evidence from the US. *J. Financ. Lit. Wellbeing* **2023**, *1*, 169–198. [\[CrossRef\]](#)
32. Mottola, G.R.; Kieffer, C.N. Understanding and using data from the National Financial Capability Study. *Fam. Consum. Sci.* **2017**, *46*, 31–39. [\[CrossRef\]](#)
33. Guzman, F.; Paswan, A.; Tripathy, N. Consumer centric antecedents to personal financial planning. *J. Consum. Mark.* **2019**, *36*, 858–868. [\[CrossRef\]](#)
34. Haryanto, J.T. Short-term versus long-term portfolio management strategies and the selection of securities. *Adv. Manag. Financ. Report.* **2024**, *2*, 11–23. [\[CrossRef\]](#)
35. Kirana, M.Y.; Havidz, S.A.H. Financial literacy and mobile payment usage as financial inclusion determinants. In Proceedings of the 2020 International Conference on Information Management and Technology (ICIMTech), Bandung, Indonesia, 13–14 August 2020. [\[CrossRef\]](#)
36. Isiaku, L.; Muhammad, A.S.; Oluwajana, D.; Kwala, A. Making mobile financial services stick: An empirical investigation into user attitudes and intentions for sustainable adoption. *J. Innov. Digit. Transform.* **2024**, *1*, 118–138. [\[CrossRef\]](#)
37. Hwang, H.; Park, H.I. The relationships of financial literacy with both financial behavior and financial well-being: Meta-analyses based on the selective literature review. *J. Consum. Aff.* **2023**, *57*, 222–244. [\[CrossRef\]](#)
38. Lusardi, A.; Mitchell, O.S. The economic importance of financial literacy: Theory and evidence. *J. Econ. Lit.* **2014**, *52*, 5–44. [\[CrossRef\]](#) [\[PubMed\]](#)
39. Cupák, A.; Fessler, P.; Hsu, J.W.; Paradowski, P.R. Confidence, financial literacy and investment in risky assets: Evidence from the Survey of Consumer Finances. In *Finance and Economics Discussion Series*; Board of Governors of the Federal Reserve System: Washington, DC, USA, 2020. [\[CrossRef\]](#)

40. Kim, K.T.; Lee, J.; Hanna, S.D. The effects of financial literacy overconfidence on the mortgage delinquency of US households. *J. Consum. Aff.* **2020**, *54*, 517–540. [[CrossRef](#)]
41. O'Brien, R.M. A caution regarding rules of thumb for variance inflation factors. *Qual. Quant.* **2007**, *41*, 673–690. [[CrossRef](#)]
42. Vatcheva, K.P.; Lee, M.J.; McCormick, J.B.; Rahbar, M.H. Multicollinearity in regression analyses conducted in epidemiologic studies. *Epidemiology* **2016**, *6*, 227. [[CrossRef](#)]

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.